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ESP32 control and protection board for a generator set

Sitex · Sousse · Jun – Jul 2025

ESP32
microcontroller

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measurements: temperature, pressure, level, current, voltage

Cloud + app
Firebase database and Android app

CONTEXT

A generator set was supervised by hand. I designed an electronic board that measures it, protects it and lets staff watch it remotely.

WHAT I DID

- Studied and chose the sensors: temperature, oil pressure, float level sensor, ACS712 Hall-effect current sensor, voltage measurement.
- Added a 5 V relay output and voltage regulation to protect the ESP32.
- Simulated the circuit in Proteus and wrote the C++ firmware.
- Sent the readings to a Firebase real-time database and built an Android app with MIT App Inventor.

TOOLS AND METHODS

ESP32 · C++ · Proteus · Firebase · MIT App Inventor · Sensors

OUTCOME

A working design that replaces manual checks with continuous measurement, remote monitoring and automatic protection.

Wassim Farhat · Electrical & Automation Engineering
wassim.farhat@polytechnicien.tn